MIL-P-48870 (PA)
1 March 1976

MILITARY SPECIFICATION

PROJECTILE, 155MM, TEST: M692

LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of Projectile, designated as Projectile, 155MM, Test: M692 with dummy wedges and live explosive charge assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1320

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

MIL-STD-286	-	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-453	-	Inspection, Radiographic

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKAGE DRAWINGS

8837839	- Pallet for Projectile, 155MM, HE, M470 or M483
9298320	- Projectile, 155MM, Test: M692
9298325	- Mine, Dummy

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

. Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Projectile.-The Projectile Assembly and Components shall comply with all requirements specified on Drawing (Dwg.) 9298320, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Ballistic Functioning.-The projectile shall function satisfactorily as defined in 6.5.1 when fired from a weapon for which it is standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

3.4 Radiographic Examination of Projectile.-The projectile shall have each layer of munitions properly positioned with respect to the keyway and shall have the column segments (in layers 1 through 4) aligned to form a straight column when the projectile is radiographically examined as specified in 4.5.4. Radiographs shall be taken in accordance with MIL-STD-453 and shall be of a quality which will permit proper evaluation as determined by the responsible Government Technical Agency.

3.5 Use of Adhesives and Sealants.-The contractor shall develop a documented operational procedure for utilization of adhesives and sealants. When applicable this procedure shall provide for quality control of raw materials and process to be used; details associated with intended use, such as proper handling, mixing, curing cycles, pot life, etc., cleaning agents, safety precautions, special ventilation, drying or other processes, and any other information of a similar nature that is considered necessary for the particular adhesive or sealant in use. This procedure shall be submitted to the Government contracting officer prior to use; and shall be updated whenever a change is made such that the original assurance of product quality is affected.

3.6 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. All required marking and stamping shall be neat and sharply defined.

3.7 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

4.2 Classification of Inspections.-The following type of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of three (3) M692 Test Projectiles as designated by the Contracting Officer.

4.3.2 Inspections to be performed.-The projectiles will be subjected by the Government to any or all of the examinations and tests specified in this specification and any or all requirements of the applicable drawing. The inspection shall include the radiograph examination specified in 4.5.4. The projectiles shall then be forwarded to a Government Proving Ground designated by the Contracting Officer for ballistic testing. The projectiles shall be tested as specified in 4.5.3 in the following zones:

a. One of the projectiles shall be fired at low zone
b. One of the projectiles shall be fired at high zone
c. One of the projectiles shall be fired at excess pressure

4.3.3 Rejection.-See MIL-A-48078.

In addition, the first article sample shall be rejected if the ballistic test results reveal any one of the following defects:

a. Any separation of metal parts occurs prior to initiation of the nose fuze.
b. Any projectile fails to function satisfactorily (See 6.5).

In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity. Until the first article quantity is accepted, the contractor is not authorized by the Government to proceed with regular production unless otherwise directed by the Contracting Officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of projectiles shall contain:

a. Projectile metal parts assemblies of one lot interfix number from one manufacturer.
b. Hardware components of one lot interfix number from one manufacturer.
c. Dummy mines of one lot interfix number from one manufacturer.
d. Charge assembly propellant from one lot only.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298340
4.4.2.1	Projectile metal parts assembly (prior to insertion of activation assembly)				NEXT HIGHER ASSEMBLY 9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Depth of cavity, min.		100%	3.2	Gage
132	Rust or foreign matter on interior wall, threads, or face at base end		0.40%	3.2	Visual
133	Base or ogive thread damaged		0.40%	3.2	Visual
134	Lubricant coating missing, incomplete, damaged, or peeling		0.40%	3.2	Visual
135	Lubricant coating present on threads		0.40%	3.2	Visual
136	Rotating band damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298395
4.4.2.2	Activation Simulator Assembly				NEXT HIGHER ASSEMBLY 9208320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Distance from top of piston to forward face of pusher plate, max.		0.40%	3.2	Gage
132	Disc pad missing or loose		0.40%	3.2	Visual
133	O-ring missing, damaged, or not lubricated		0.40%	3.2	Visual
134	Activation simulator rod bent		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4910

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298325
4.4.2.3	Mine, Dummy (Prior to radial compression)				NEXT HIGHER ASSEMBLY 9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Mine damaged		0.65%	3.2	
202	Paint missing, wrong color, or on incorrect surfaces		0.65%	3.2	
203	Evidence of poor workmanship		0.65%	3.6	

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298320
4.4.2.4	Wedge layer (prior to assembly into stack-up)				NEXT HIGHER ASSEMBLY 9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Layer does not meet specified loads		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298320
4.4.2.5	Mine Stack-up (After radial compression and prior to assembly into projectile)				NEXT HIGHER ASSEMBLY 9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Any layer not radially sized		0.40%	3.2	Certification
102	Any part missing		0.40%	3.2	Visual
103	Any layer not properly aligned with keyway		0.40%	3.2	Visual
Minor					
201	Column lengths not equalized (within tolerance)		0.40%	3.2	Certification
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Projectile (After assembly of dummy mines and prior to assembly of base plug)				9298320
					NEXT HIGHER ASSEMBLY
					9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Base clearance inadequate		0.40%	3.2	Gage
102	Base pad missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298335
4.4.2.7	Base plug (prior to assembly to body)				NEXT HIGHER ASSEMBLY 9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Minor					
201	Sealant not applied to threads		0.65%	3.2	Visual
202	Threads damaged		0.65%	3.2	Visual
203	Nicks on shoulder adjacent to threads		0.65%	3.2	Visual
204	Seal ring missing or not lubricated		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Projectile (after assembly of base and prior to assembly of ogive)		9298320
			NEXT HIGHER ASSEMBLY 9298320

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Base turns when specified disassembly torque is applied		100%	3.2	Gage
102	Excessive gap between body and base		100%	3.2	Gage
Minor					
201	Sealant on exterior surface		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298339
4.4.2.9	Shim, Ogive (prior to assembly to projectile)				NEXT HIGHER ASSEMBLY 9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B 131	Thickness incorrect for required preload		100%	3.2	Gage

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Ogive (prior to assembly to projectile)				9298320
					NEXT HIGHER ASSEMBLY
					9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Compound not applied to external threads		0.65%	3.2	Visual
202	External threads damaged		0.65%	3.2	Visual
203	Nicks on shoulder adjacent to external threads		0.65%	3.2	Visual
204	Seal ring missing or not lubricated		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298320
4.4.2.11	Projectile (immediately after assembly of ogive and prior to assembly of charge and sleeve assembly)				NEXT HIGHER ASSEMBLY 9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Gap between body and ogive beyond specified limits		100%	3.2	Gage
102	Concentricity of ogive to bourrelets (2 places)		100%	3.2	Gage
103	Radiographic examination		100%	3.4	4.5.4
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298370
4.4.2.12	Spotting, Charge Assembly				NEXT HIGHER ASSEMBLY 9298360
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cup not completely filled (settling permitted)		0.40%	3.2	Visual
102	Tape missing or loose		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298360
4.4.2.13	Charge and Sleeve Assembly (prior to insertion of charge assembly)				NEXT HIGHER ASSEMBLY 9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Spotting charge assembly not secure (when applicable)		0.40%	3.2	Visual/Manual
102	Closure cup missing (when applicable)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Charge Assembly (prior to insertion into charge and sleeve assembly)				9298363
					NEXT HIGHER ASSEMBLY
					9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Weight of propellant incorrect		0.40%	3.2	Balance
102	Bag waterproofness destroyed (separation of heat seal, bag punctured etc)		0.40%	3.2	Visual
103	Leakage test	32-0-1		3.2	4.5.2
104	Moisture and Volatiles content of propellant			3.2	4.5.1
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Charge and Sleeve Assembly (prior to insertion into projectile)				9298360
					NEXT HIGHER ASSEMBLY 9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Staking of charge cover missing or inadequate (4)		0.40%	3.2	Visual
132	Cover damaged (dented, punctured etc)		0.40%	3.2	Visual
133	Charge missing (when applicable)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Projectile (after assembly of charge and sleeve assembly and prior to assembly of lifting plug)	1 OF 1	9298320
			NEXT HIGHER ASSEMBLY 9298320

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Weight		100%	3.2	4.5.5
Major B					
131	Retaining ring missing or improperly assembled		0.40%	3.2	Visual
132	Foreign matter in fuze cavity		0.40%	3.2	Visual
133	Lifting plug gasket missing		0.40%	3.2	Visual
134	Lifting plug threads not greased		0.40%	3.2	Visual
135	Threads damaged		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298320
4.4.2.17	Projectile (prior to assembly of grommet)				NEXT HIGHER ASSEMBLY 9298320
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Lifting plug turns when disassembly torque is applied		0.40%	3.2	Test
102	Diameter of bourrelet, max (applicable after retouching paint)		0.40%	3.2	Gage
103	Rotating band damaged		0.40%	3.2	Visual
104	Obturator missing or damaged		0.40%	3.2	Visual
Minor					
201	Protective coating with a total of bare areas in excess of 1/4 square inch		0.65%	3.2	Visual
202	Marking missing or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8837839
4.4.2.18	Pallet (loaded)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Projectile lifting plug missing or damaged		0.40%	3.2	Visual
132	Projectile grommet assembly missing, damaged, or improperly assembled		0.40%	3.2	Visual
133	Board broken or split		0.40%	3.2	Visual
Minor					
201	Strap width incorrect		0.65%	3.2	Visual
202	Strapping improperly assembled		0.65%	3.2	Visual
203	Marking missing, incorrect or unidentifiable		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-48870 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Wood box (prior to closing) (Applicable when projectiles are not shipped on pallets)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Provisional pack warning sheet missing			3.2	Visual
102	Projectile not secured			3.2	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.20	Wood box (after closing and sealing) (applicable when projectiles are not shipped on pallets)		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Provisional pack warning missing			3.2	Visual
102	Exterior markings missing, incorrect or unidentifiable			3.2	Visual
103	Cover not secured			3.2	Visual
Minor					
201	Box damaged			3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

4.4.3 Testing.-Testing is described in First Article Inspection and Quality Conformance Inspection Tables.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See MIL-A-48078.

4.5 Test Methods and Procedures.

4.5.1 Moisture and Total Volatiles Content of Charge Assembly.-The contractor shall provide adequate controls to insure compliance with the applicable requirement. The moisture and total volatiles content of each lot of propellant shall be determined at the loading station at the time of loading the charge assemblies. This determination shall be made at least once for eight (8) hours of production. Sampling and testing of the propellant shall be in accordance with MIL-STD-286, Method 101.1.2. If the moisture and total volatiles content fails to comply with the requirement specified on the applicable drawing and loading of the charge assemblies has not begun, the propellant lot shall be rejected until corrective action has been taken as verified by repeating the moisture and total volatiles content determination. If loading has begun, charge assemblies loaded with the non-conforming propellant shall be rejected.

4.5.2 Leakage Test of Charge Assembly.-The charge assemblie shall be fully immersed in water and subjected the pressure specified on the applicable drawing using equipment as specified in 4.4.4. Any assembly which emits bubbles shall be rejected. Charge assemblies so tested will not be returned to any lot. (Destructive test).

4.5.3 Ballistic Functioning.-This test shall be conducted at a Government proving ground in accordance with approved Acceptance Test Procedures. The round shall be fitted with a fuze which is standard for the projectile. The projectile fuze setting and the quadrant elevation of the weapon shall be adjuste to achieve a burst height between fifteen hundred (1500) and twenty five hundred (2500) feet over a prepared impact area. A photograph shall be taken of the fuze setting for each round

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

immediately prior to firing and each photograph shall be marked for future identification with the round. The sample projectile(s) shall be tested in the specified zone(s) using weapon, equipment, methods and procedures in accordance with the acceptance test procedures referenced above. The M109A1 weapon shall be used unless otherwise specified by the procuring agency.

4.5.3.1 Low Zone.-The sample projectile to be subjected to this test shall be fired using the propelling charge for zone 1 in the M109A1 weapon. Observation shall be made for failure to comply with all the requirements of 3.3.

4.5.3.2 High Zone.-The sample projectile to be subjected to this test shall be fired using the propelling charge for zone 8 in the M109A1 weapon. Observation shall be made for failure to comply with all the requirements of 3.3.

4.5.3.3 Excess Pressure.-The sample projectile to be subjected to this test shall be fired using a propelling charge adjusted to produce the required pressure. (See 6.5.2) Observation shall be made for failure to comply with the requirements of 3.3.

4.5.4 Radiographic Examination of Projectile.-Prior to assembly of the base plug, the location of the keyway shall be marked on the exterior of each projectile. The projectiles shall be radiographed after assembly of the ogive and prior to assembly of the charge and sleeve assembly. In all exposures, the radiographic beam shall be focused along the center axis of the projectile and parallel to the face of the shelf plate. One radiograph shall be taken 10° clockwise from the keyway position, such that the column segments are in line with the source. A second radiograph shall be taken at a location with the projectile rotated 45° clockwise from the view of the first radiograph. The radiographs shall, as a minimum, show the entire cargo pack including the base and forward plate. If necessary, more than one radiograph in a view, shall be taken as required to avoid detrimental parallax effects. The radiographs shall be examined for any of the conditions specified in 3.4. All projectiles failing to conform to the requirements shall be suitably identified and removed from the lot. Projectiles and radiographs shall be identified to permit future reference. The radiograph for the ballistic sample shall be furnished with the projectiles to the Government inspector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

NOTE: Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. In addition the photographic density of the film used shall be of a quality which will produce a sharp definition of internal structure.

4.5.5 Projectile Weight Determination.-The projectile, without lifting plug, gasket and grommet assembly shall be weighed using equipment of suitable range and accuracy. The weight shall be marked in the specified location on the projectile and this marking shall be verified as being the same as the weight indicated by the equipment.

4.5.6 Process Control.-At random intervals, the Government inspector shall monitor the contractor's assembly operations to assure compliance with the requirements specified on the applicable drawings and in approved loading and assembling procedures. As a minimum, the following shall be monitored:

a. Application of pads to munitions.
b. Equalization of column segment heights by selective assembly or use of column shims.
c. Radial compression of layers prior to assembly into stack-up.
d. Loading size ring diameter.
e. Determination of ogive shim thickness including application of specified load.
f. Torquing of the ogive.
g. Axial alignment of column segment wedges in layers 1 through 4.

5. PREPARATION FOR DELIVERY

5.1 Packing (Level A).-The projectiles shall be packed in accordance with Dwg. 8837839.

5.2 Marking.-Marking shall be in accordance with Dwg. 8837839 and Code of Federal Regulations, Title 49, Parts 100-199.

5.3 Shipping.-When projectiles from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended Use.-This specification is intended for use in loading, assembling and packing the M692 test projectile.

6.2 Ordering data.-See MIL-A-48078.

6.3 Submission of Designs for Approval.-See MIL-A-48078. Submit equipment designs, as required, to Commander, Picatinny Arsenal, Attn: SARPA-QA-T-T, Dover, New Jersey 07801.

6.4 Submission of Results of Contractor Conducted Examinations and Tests.-Data shall be submitted in accordance with data item DI-R-1721 on the DD Form 1423 for the contract.

6.5 Projectile Functioning

6.5.1 Functioning at all Zones.-Satisfactory functioning of the projectile when fired at any zone, is defined as successfull delivery of the cargo to the target area with all projectile hardware remaining intact until initiation of the time fuze at a pre-determined time after firing, and also includes expulsion of the entire cargo.

6.5.2 Excess Pressure.-Excess pressure in the M109A1 weapon system is herein considered to be a chamber pressure between 29,000 psi minimum and 32,000 psi maximum.

6.6 Submission of Test Data.-In addition to the normal distribution of records, when the projectiles are procured by the U.S. Army Armament Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, Picatinny Arsenal, Attn: SARPA-QA-A-S.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1320-A319

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:42Z
Check the source to verify that this is the current version before use.